

Weekly Progress Report

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Week 4 Progress Update:

(9/8/2026 - 9/15/2026)

This Week's Focus

Reviewed literature for environmental factors to include in the species distribution model.

Candidate Variables

The following variables are currently selected based off representation in literature, percieved data availability, or in an effort to maintain cross comparison compatibility with the HSI derived in Geselbracht et al. (2024).

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Reason	Reference	Data Source
Salinity	Widely available, large driver of ecosystem suitability	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012; Barnes et al. 2007)	Water Quality Portal (DeCicco et al. 2026)
Water temperature	Widely available, large driver of ecosystem suitability	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012; Barnes et al. 2007)	Water Quality Portal (DeCicco et al. 2026)

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Variable	Reason	Reference	Data Source
Substrate / bottom type	suitable substrate required for recruitment	(lePeyre2021? ; Geselbracht et al. 2024; Barnes et al. 2007)	TBD
Water depth / bathymetry	Depth is a key driver in oyster habitat extent	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012; Barnes et al. 2007)	National Centers for Environmental Information (National Centers for Environmental Information 2018)
Dissolved oxygen	Low DO can cause oyster stress and mortality	(lePeyre2021? ; Geselbracht et al. 2024; Pollack et al. 2012)	Water Quality Portal (DeCicco et al. 2026)
Chlorophyll- a	Common proxy for food resources supporting oyster growth	(lePeyre2021? ; Geselbracht et al. 2024)	Water Quality Portal (DeCicco et al. 2026)
Oyster bed locations	important for SDM modeling requiring species presence data	(lePeyre2021? ; Geselbracht et al. 2024)	Florida Fish and Wildlife Conservation Commission (Florida Fish and Wildlife Conservation Commission 2015)

References

- Barnes, T. K., A. K. Voley, K. Chartier, F. J. Mazzotti, and L. Pearlstine. 2007. “A Habitat Suitability Index Model for the Eastern Oyster (*Crassostrea virginica*), a Tool for Restoration of the Caloosahatchee Estuary, Florida.” *Journal of Shellfish Research* 26 (4): 949–59. [https://doi.org/10.2983/0730-8000\(2007\)26%5B949:AHSIMF%5D2.0.CO;2](https://doi.org/10.2983/0730-8000(2007)26%5B949:AHSIMF%5D2.0.CO;2).
- DeCicco, Laura, Robert Hirsch, David Lorenz, Jordan Read, Jordan Walker, Lindsay Platt, David Watkins, et al. 2026. *dataRetrieval: R Packages for Discovering and Retrieving Water Data Available from u.s. Federal Hydrologic Web Services* (version 2.7.25). Reston, VA: U.S. Geological Survey; U.S. Geological Survey. <https://doi.org/10.5066/P9X4L3GE>.
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- Geselbracht, L., M. Johnston, B. M. DeAngelis, and A. Birch. 2024. “Estuary-Specific and Adaptive Habitat Suitability Index Model for the Eastern Oyster *Crassostrea virginica* in the Pensacola Bay System, Florida, USA.” *Coastal Management* 52: 17–34. <https://doi.org/10.1080/08920753.2024.2335129>.

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- Pollack, Jennifer Beseres, Andrew Cleveland, Terence A. Palmer, Anthony S. Reisinger, and Paul A. Montagna. 2012. “A Restoration Suitability Index Model for the Eastern Oyster (*Crassostrea Virginica*) in the Mission-Aransas Estuary, TX, USA.” *PLoS ONE* 7 (July). <https://doi.org/10.1371/journal.pone.0040839>.